

Design Verification Plan and Report (virtual) - GridStore-D3

VBF-T2R1-DVPR-01 | t2_r1 grid energy storage | 2026-08-25 | values mechanically sourced from parameter set / simulation outputs

Verification items (virtual-test version)

#	Item	Condition	Result value	Determination	Source
1	1C discharge capacity	1C_discharge, 298.15 K	5.06626 Ah (nominal 5.0 Ah)	PASS (>= nominal)	cell/r6_d3_1c.json:capacity_ah
2	Energy density	contract caliber (calc-energy)	465.62 Wh/kg	PASS vs >= 327.18	cell/r6_d3_energy.json:energy_density_wh_kg
3	4C fast-charge plating	4C_charge_45C --plating, 318.15 K	anode pot min +0.0498 V	PASS (min >= 0)	cell/r6_d3_4c.json:anode_potential_v
4	4C temperature rise	4C_charge_45C --thermal lumped	T_max 342.39 K (+24.2 K)	informational (no contract red line)	cell/r6_d3_4c.json:T_max_K
5	Voltage window	parameter-set limits	2.5 - 4.2 V	PASS (as specified)	Chen2020 dump
6	-20C discharge retention	lowT_discharge, 253.15 K	99.571% (5.04452/5.06626 Ah)	PASS vs >= 90%	cell/r6_d3_lowT_ret.json
7	-20C true-soak cross-check	lowT + Initial temperature 253.15 K	99.571% (5.04452 Ah), T_max 266.71 K	PASS vs >= 90%	cell/r6_d3_lowT_soak.json
8	SEI @100 cyc	aging_1C_100cyc (1C CC 2.5-4.2 V)	9.087 nm	PASS vs <= 500 nm	cell/r6_d3_aging100.json:sei_thickness_nm_end
9	SEI @500 cyc	aging --cycles 500	25.317 nm	PASS vs <= 550 nm	cell/r6_d3_aging500.json:sei_thickness_nm_end
10	Cycle-capacity trajectory shape	aging protocol	flat (0.3032->0.3041 Ah, no climb artifact)	informational	cell/r6_d3_aging500.json:capacity_ah_per_cycle

Items marked N/A (beyond pure simulation boundary / requires physical experiment)

Item	Reason
Nail penetration	N/A (requires physical experiment)
Overcharge to thermal runaway	N/A (protocol available but not in task contract; not executed)
Crush, drop	N/A (requires physical experiment)
Rate-pulse internal resistance	N/A (calc-energy DCR scalar is a step-density artifact - design_spec section 6)
Real-electrode coating validation (k_sei 1e-16 bridge)	N/A (requires Stage 2 material work / physical validation - honest gap)
CAD structure model	N/A (not requested in task clarification; optional deliverable)

Conclusion: 5/5 contractual verification items PASS with margin (ED x1.42, plating-free at 4C, SEI 9.1/25.3 nm vs 500/550 nm, -20C retention 99.6%). Uncovered items cited directly as paper limitations. Known tool artifacts annotated (aging per-cycle-capacity variable freezes at first-discharge value while real steps are full 1C swings - probe-grounded; calc-energy midpoint/DCR step-density artifacts).